

CARLTON FL: Federated Deep Reinforcement Learning for Distributed Channel Allocation in Cognitive Interference Networks

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Abstract—We consider distributed dynamic channel allocation (DCA) in cognitive interference networks, where a shared band is divided into K non-orthogonal channels and inter-carrier interference (ICI) couples the decisions of spatially overlapping networks. The task is closed-loop and multi-agent: each network manager must perceive local SINR quality, retain experience over time, and select channels under partial observability, while coordinating with peers that may not share raw sensing data. State-of-the-art deep reinforcement learning (DRL) solutions for this setting operate under the *centralized training with decentralized execution* (CTDE) paradigm of CARLTON [1]: although execution is distributed, training requires every network to export its raw sensing history to a shared replay memory and concentrates all gradient computation on a single learner. We propose CARLTON FL, a federated DRL framework that removes the centralized learner. Each network manager acts as a federated client that owns a private, persistent replay buffer (agent memory), performs E local update steps of a target-network-free mellowmax value learner, and uploads only model parameters, which a server aggregates by FEDAVG or FEDPROX. We identify three sources of statistical heterogeneity specific to federated DRL for DCA—topological asymmetry, policy-induced non-stationarity, and inter-agent coupling—and show that the FEDPROX proximal term keeps long local phases stable under them. On the operational channel-quality metrics of the centralized state of the art, evaluated over 420 games per method, the PyTorch CTDE baseline (CARLTON) remains slightly ahead: all-sample composite $\mathbb{E}[(CQ + \min CQ)/2]$ of 0.909 versus 0.905 for FEDPROX and 0.902 for FEDAVG, with weighted scores near 0.931 for all three. In-sample ($N \leq 7$) the gap shrinks to within metric resolution (0.983 CTDE, 0.982 FEDAVG, 0.978 FEDPROX). Thus federation does *not* outperform CTDE; it trades a thin margin for a strictly stronger privacy constraint in which raw sensing data never leave the network that produced them.

Index Terms—Agentic AI, multi-agent deep reinforcement learning, federated learning, distributed dynamic channel allocation (DCA), privacy-preserving edge intelligence, cognitive interference networks, inter-carrier interference (ICI), SINR-aware resource management.

I. INTRODUCTION

THE growing demand for wireless connectivity, together with spectrum scarcity, has made *dynamic channel al-*

location (DCA) a central mechanism in emerging cognitive communication systems. In the 6G edge regime, independently managed networks must operate as closed-loop agents: they perceive local spectrum quality, retain operational memory across time, reason about channel decisions under uncertainty, and coordinate with neighbors under strict latency and privacy constraints—capabilities that go beyond isolated prediction or one-shot optimization. In contrast to static allocation, DCA lets every network revise its operating channel in response to the interference it currently experiences. The mechanism is relevant across cognitive radio networks, 6G in-X sub-networks, vehicular ad-hoc networks, wireless sensor and IoT networks, and wireless mesh networks—all characterized by dense coexistence of independently managed networks within a shared band [1].

Most existing formulations simplify the problem by assuming perfectly orthogonal channels, so that interference occurs only between networks occupying the *same* channel. This assumption rarely holds in practice. Spectral leakage and hardware imperfections produce *inter-carrier interference* (ICI), and partially overlapping channels measurably affect network performance [2]. Once orthogonality is dropped, a channel is interfered with not only by co-channel transmissions but also, to a degree decaying with spectral distance, by every neighboring channel. The resulting coupling is continuous rather than binary, varies over the network graph with geographic distance, and inflates the search space to the point where classical allocation algorithms become inefficient.

Four properties make distributed DCA under ICI particularly difficult. First, each network manager observes only its own sensing measurements and therefore holds a strictly partial view of the global interference state, placing the problem in the class of partially observable Markov decision processes. Second, the environment is non-stationary: when one manager switches channel, the observation and reward distributions of all of its neighbors change simultaneously. Third, the optimal joint assignment generalizes graph coloring to weighted, non-binary interference and is intractable [3]; it cannot be recomputed centrally at the timescale on which topologies evolve. Fourth, spectrum mobility is itself costly, since every reallocation interrupts service, so a useful policy must converge quickly and then remain stable.

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A. DRL Algorithms for DCA

Multi-agent DRL is an attractive framework for DCA precisely because it requires no explicit model of the interference environment: agents learn a channel-selection policy directly from interaction and implicitly discover the spatial reuse structure of the network. Following the early multi-user DRL work of Naparstek and Cohen [4], a substantial line of research has applied deep Q -learning and its variants to spectrum access [5], [6], [7]. The current state of the art for the SINR-aware setting considered here is CARLTON [1], which reports gains of roughly 45% over a random-agent benchmark [12] and 20% over the Jamming Avoidance Response (JAR) heuristic used in [1], outperforms the DRL baselines MADDQN-DCA [5] and IDSA [6], and comes within about 2.5% of a centralized graph-coloring upper bound. It is the benchmark against which we position our framework; our goal is a privacy-preserving federated counterpart that approaches—not exceeds—this CTDE reference.

What all of these methods share—and what this paper targets—is their *training* architecture. Although the learned policies execute in a fully distributed manner, with no controller and no message exchange between networks at test time, they are trained under the CTDE paradigm, illustrated in Fig. 1(a): during an episode each network manager n writes its transitions into a small local memory $\mathcal{D}^n$ holding its T decision points; at the end of the episode every $\mathcal{D}^n$ is flushed into a global replay memory (GRM) of capacity S_z , and one central learner draws mini-batches from the GRM and performs all gradient updates. The local buffers are mere staging areas; the learning signal is entirely centralized.

This imposes three limitations that are decisive in practice. *Privacy and data sovereignty*: every transition contains the agent’s quality vector, which encodes fine-grained information about its spectral environment, its interference neighborhood, and—by implication—its physical location, its number of active users, and its traffic behavior; exporting it is unacceptable when the participating networks belong to competing operators, distinct tactical units, or different regulatory domains. *Computational centralization*: all gradient computation is performed by one node, which becomes both a scalability bottleneck and a single point of failure, while the network managers—which the system model already assumes to be capable computing nodes—remain idle. *Deployment realism*: independently operated networks rarely accept a common trusted training authority. There is a genuine tension here, since the case for distributed DCA is made precisely on the grounds of communication overhead and computational complexity [1], yet those same costs reappear in the training phase.

Federated learning [13], [14] is the natural answer, but transplanting it into multi-agent DRL for DCA is not mechanical. In supervised federated learning the client datasets are fixed and non-IID only through their label distributions. Here each client *generates* its own data with the policy it is currently learning, inside an environment whose other occupants are the remaining clients. The heterogeneity is consequently structural, non-stationary, and coupled—the regime in which plain parameter

averaging is known to be fragile.

B. Main Results

We develop CARLTON FL, a federated DRL framework for SINR-aware distributed DCA under ICI, shown in Fig. 1(b). The global replay memory is eliminated: each network manager is a federated client owning a *persistent* private buffer, retained across episodes rather than flushed to a central store. A round proceeds in four phases—the server broadcasts w^t ; each client plays one interaction episode; each client performs E local gradient steps on samples drawn exclusively from its own buffer; each client uploads only w_n^{t+1} , which the server aggregates. Raw experience never leaves the client, and once training terminates the server is no longer needed. Our contributions are as follows.

C1. A federated DRL framework for SINR-aware DCA under ICI. To the best of our knowledge, this is the first federated reinforcement learning framework for distributed channel allocation in cognitive interference networks with non-orthogonal channels. We show that the episodic transfer of experience into a global replay memory—the mechanism on which the centralized state of the art relies—can be replaced by persistent per-agent buffers with parameter-level aggregation, without altering the observation space, the action space, or the message-free execution phase. We release a complete open-source implementation together with a faithful reproduction of the centralized baseline, so that the two paradigms differ only in how learning is coordinated.

C2. Taming non-IID multi-agent experience with FEDPROX. We characterize three sources of statistical heterogeneity specific to federated DRL for DCA—topological and load asymmetry across agents, policy-induced non-stationarity of each local buffer, and inter-agent coupling of the local objectives—and show that together they make client drift the dominant failure mode when the number of local steps is large. We adopt the FEDPROX proximal objective and conduct a systematic sweep over the local-epoch count E and the proximal coefficient μ , identifying $E^* = 10$ and $\mu^* = 0.01$ as the configuration that best balances communication efficiency against stability.

C3. Privacy-preserving training that remains competitive with—but does not surpass—CTDE. All SINR measurements, quality vectors, user counts, and transition tuples remain in the network that generated them; only model parameters cross the network boundary, and the training computation is distributed across the network managers. Federated clients train under a stricter information constraint ($\rho = 1$, no social term) than CARLTON ($\rho = 0.7$). On the primary channel-quality metrics over 420 games, PyTorch CTDE leads slightly (all-sample composite 0.909, in-sample 0.983) while FEDPROX / FEDAVG attain 0.905 / 0.902 all-sample and 0.978 / 0.982 in-sample. The DeepMellow training reward is reported only as a learning diagnostic; the claim of this paper is competitiveness under privacy, not superiority over CARLTON.

C4. Agentic multi-agent DCA with private memory and communication-efficient coordination. We cast each network manager as an autonomous agent with the four closed-loop

TABLE I
ATTENUATION $T(k, \tilde{k})$ WITH RESPECT TO SPECTRAL DISTANCE, WHERE
 f_k AND $f_{\tilde{k}}$ ARE THE CARRIER FREQUENCIES OF CHANNELS k AND $\tilde{k}$,
RESPECTIVELY [10].

Spectral distance $ k - \tilde{k} $	Attenuation (dBm)
0	0
1	20
2	40
3	50
4	60
≥ 5 and $ f_k - f_{\tilde{k}} /f_k \leq 0.05$	95
else	110

capabilities required for intelligent networks: *perception* of wideband BSINR/quality vectors, *memory* via a persistent private replay buffer retained across episodes, *reasoning* via a mellowmax value policy under partial observability, and *coordination* via parameter-only federation rather than raw experience pooling. This yields a trustworthy, privacy-preserving form of multi-agent edge intelligence for SINR-aware resource management, aligned with communication-efficient distributed learning under non-IID local experience.

The remainder of this paper is organized as follows. Section II formalizes the system model and the constrained DCA objective, and maps them onto a federated optimization problem. Section III presents the CARLTONFL algorithm. Section IV reports the experimental study, and Section V concludes.

II. SYSTEM MODEL AND PROBLEM FORMULATION

A. Description of the System

We consider N coexisting wireless networks $\mathcal{N} = \{\mathcal{N}^1, \dots, \mathcal{N}^N\}$, where $\mathcal{N}^n$ serves $M^n \in \{1, \dots, M_{\max}\}$ users. The total bandwidth is divided into K equally sized channels of bandwidth B , indexed by $\mathcal{K} = \{1, \dots, K\}$ with carrier frequencies $\{f_1, \dots, f_K\}$, and each network transmits on exactly one channel per time slot.

Crucially, the channels are *not* orthogonal. Spectral leakage and practical filter roll-off cause every transmission to deposit energy in neighboring channels, so a network operating on channel $\tilde{k}$ interferes with a receiver tuned to k by an amount that decays with the spectral distance $|k - \tilde{k}|$ rather than vanishing for $k \neq \tilde{k}$ (Fig. 2). The residual coupling is captured by an attenuation function $T(k, \tilde{k})$, listed in Table I, which decays with the spectral distance $|k - \tilde{k}|$ and saturates beyond four channels of separation. This turns allocation into a weighted rather than binary conflict problem and substantially enlarges the search space. Throughout we write $(\cdot)^{l,n}_{m,j,k}$ for a quantity associated with the user pair $m \in \mathcal{N}^l$, $j \in \mathcal{N}^n$ on channel k ; intra-network interference is assumed resolved by the medium access control scheduler.

Propagation follows the Egli model [11]. For users $i, j \in \mathcal{N}^n$ with i transmitting to j on channel k , the received SINR is

$$\text{SINR}_{i,j,k}^{n,n} = \frac{PR_{i,j,k}^{n,n}}{I_T + I_{j,k}^{n,n}}, \quad (1)$$

$$PR_{i,j,k}^{n,n} = PT_{j,k}^n - PL_{i,j,k}^{n,n} - \chi_{i,j,k}^{n,n} - \psi_{i,j,k}^{n,n},$$

where I_T is thermal noise, $I_{j,k}^n$ the aggregate interference from $\mathcal{N} \setminus \mathcal{N}^n$, PT the transmit power in dBW, PL the path loss, and χ, ψ the fading and shadowing terms. The Egli path loss is

$$PL_{i,j,k}^{n,n} = 40 \log_{10}(d_{i,j}^{n,n}) - 20 \log_{10}\left(\frac{40}{f_k}\right) - 20 \log_{10}(H_{T_i}^n H_{R_j}^n) - 10 \log_{10}(G_{T_i}^n G_{R_j}^n) \text{ [dB]}, \quad (2)$$

with d the distance in meters, f_k the carrier in MHz, H_T, H_R the antenna heights and G_T, G_R the gains. Interference aggregates over all other networks and their users,

$$I_{j,k}^n = \sum_{l \in \mathcal{N} \setminus \{n\}} \sum_{m=1}^{M^l} \phi_{j,m,k}^{n,l}, \quad \phi_{j,m,k}^{n,l} = PR_{j,m,k}^{n,l} - T(k, \tilde{k}), \quad (3)$$

where $\tilde{k}$ is the channel occupied by the interferer. Equation (3) is where non-orthogonality enters: for $\tilde{k} = k$ we have $T = 0$ and the interferer is felt at full strength, while for $|k - \tilde{k}| \geq 1$ it is attenuated but not eliminated. The noise floor is $I_T = 10 \log_{10}(k_B \text{Temp } B) + 10 \log_{10}(\text{NF}) + 30 \text{ dBm}$, equal to -104.9 dBm at room temperature with $B = 2 \text{ MHz}$ and a 6 dB noise figure.

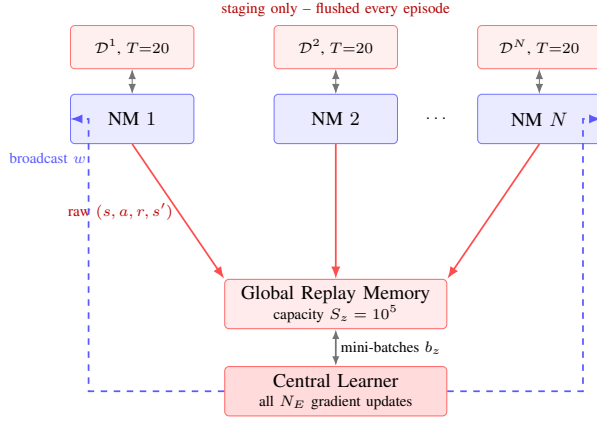
Scenario generation and the network manager. The number of networks N is sampled uniformly, and each M^n is drawn uniformly from $\{1, \dots, M_{\max}\}$; users scatter around their network center by a bivariate Gaussian with covariance $\text{diag}(50^2, 50^2) \text{ m}^2$. Centers are placed sequentially: the first uniformly over a square whose extent grows with N , and every subsequent center attached to a randomly chosen existing center at a random radius $r \sim U[x_1, x_2]$ and angle $\theta \sim U[0, 2\pi]$.¹ This produces clustered, partially overlapping deployments rather than a uniform scatter, which is what makes the interference graph non-trivial. Within each network the user minimizing the total distance to all others is designated the *network manager*, mirroring the centralized-unit role in 5G local networks: it gathers the per-user reports over a dedicated control channel, selects the channel, and disseminates the decision. Fig. 3 shows a representative instance. Decisions are sequential: each network is assigned a serial number fixing its turn, which emulates propagation and processing delays and prevents simultaneous conflicting decisions. The horizon is TN steps, so every network receives exactly T decision points regardless of N .

B. The Objective

Let $c^n \in \mathcal{K}$ denote the channel of $\mathcal{N}^n$. Averaging (1) over the messages a user receives from the other $M^n - 1$ users, and then over users, yields the network-level quantity $\overline{\text{SINR}}_{c^n}^n$. The allocation goal is to maximize the mean network SINR subject to a target QoS level for *every* participating network,

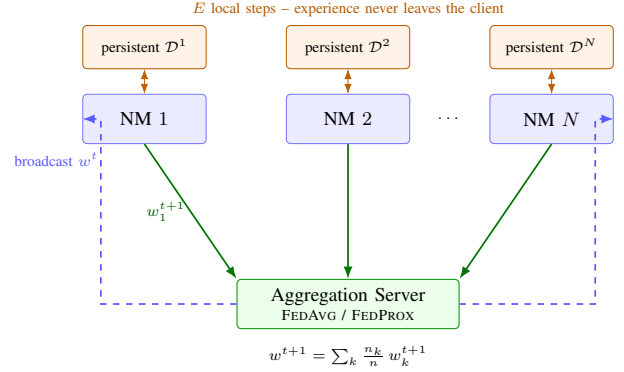
$$\max_{\{c^n\}} \frac{1}{N} \sum_{n=1}^N \overline{\text{SINR}}_{c^n}^n \quad \text{s.t.} \quad \overline{\text{SINR}}_{c^n}^n \geq \text{SINR}^*, \quad \forall n. \quad (4)$$

¹The reference implementation released with [1] draws r and θ with an integer-valued sampler, restricting θ to six discrete angles. We corrected this in both our framework and the reproduced centralized baseline, so that all methods are trained and evaluated on identically distributed topologies.



(a) Centralized training (CTDE) – state of the art

raw sensing data leaves every network • single point of failure • all computation on one node



(b) CARLTON FL (proposed)

only parameters are exchanged • computation distributed over clients • FEDPROX limits client drift

Fig. 1. Training paradigms for distributed DCA. (a) The centralized state of the art flushes the local memories $\mathcal{D}^n$ of all network managers (NM) into a global replay memory, from which a single learner performs every gradient update. (b) In CARLTON FL each NM keeps a *persistent* private buffer, performs E local updates on it, and uploads only parameters. The execution phase is message-free and identical in both cases.

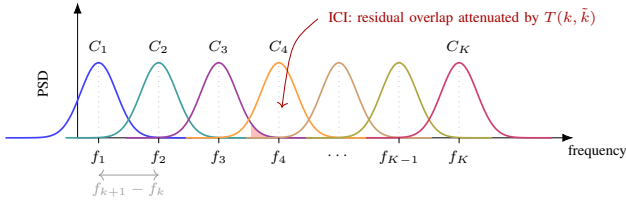


Fig. 2. Non-orthogonal channel partition: adjacent channels overlap because of spectral leakage, so a transmission on k reaches a receiver tuned to k attenuated by $T(k, \bar{k})$ rather than suppressed.

The constraint is essential: maximizing the average alone admits solutions in which a few well-placed networks capture most of the spectrum while others are starved.

Solving (4) directly would require each manager to collect real-valued SINR measurements from all of its users on all K channels, which is prohibitive for large networks and battery-limited terminals. Each user therefore reports a single bit per channel, $\text{BSINR}_{j,k}^n = 1(\text{SINR}_{j,k}^n > \text{SINR}_T)$, and the manager averages the resulting $M^n \times K$ matrix along the user axis to obtain the *quality vector*

$$\text{QV}_k^n = \frac{1}{M^n} \sum_{j=1}^{M^n} \text{BSINR}_{j,k}^n \in [0, 1]. \quad (5)$$

Entry QV_k^n is the fraction of users of $\mathcal{N}^n$ that would meet the QoS target on channel k ; averaging removes the dependence on network size. Under this communication constraint the objective becomes $\max_{\{c^n\}} \frac{1}{N} \sum_n \text{BSINR}_{c^n}^n$ subject to the same per-network floor. The quality vector is the only environmental information available to the agent—and, in the centralized paradigm, the quantity that must be shipped to the global replay memory. Measuring SINR across all channels presupposes wideband sensing, which modern software-defined radio front ends make practical [15].

C. The Federated Learning Problem

Each network manager is a computing node that privately holds all data generated by its own network: the per-user BSINR reports, the quality vectors $\text{QV}^n(t)$, the channels it selected, and the rewards it received. Identifying the manager of $\mathcal{N}^n$ with federated client n therefore requires no additional infrastructure assumption—it merely refrains from exporting data the manager already possesses. Writing $F_n(w)$ for the expected temporal-difference loss incurred by parameters w on the transitions stored at client n , and n_k for the number of samples it holds, learning becomes the federated problem

$$\min_w F(w) = \sum_{k=1}^N \frac{n_k}{\sum_j n_j} F_k(w), \quad (6)$$

whereas CTDE training minimizes a single empirical loss over the union of all clients' transitions. The two coincide only when the local distributions are identical, which here they demonstrably are not.

Three structural properties make the local objectives genuinely heterogeneous, and all three are visible in Fig. 3. First, M^n varies over an order of magnitude across networks, and since (5) averages over M^n users, quality vectors of small and large networks differ in granularity and noise. Second, the number of networks within the neighbor threshold Γ differs across agents, so agents at the center of a cluster face dense interference while peripheral agents see an almost empty spectrum. Third, because a network's observation depends on the channels currently occupied by its neighbors, each client's distribution is continuously reshaped by the concurrent policy updates of the others, making the local objectives both non-stationary and mutually coupled.

III. THE CARLTON FL ALGORITHM

A. Background: Value-Based DRL without a Target Network

Each network manager is an autonomous agent that observes its environment, selects a channel, and receives a scalar reward.

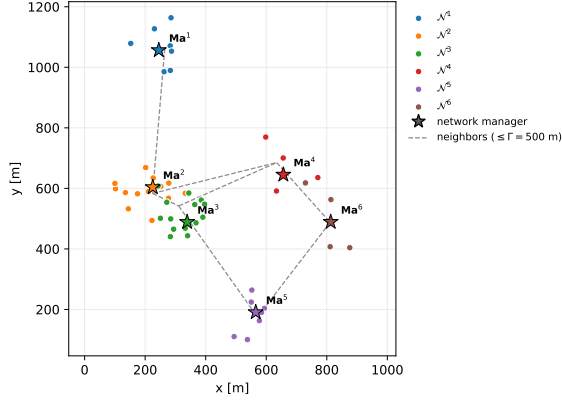


Fig. 3. Representative six-network scenario produced by our simulator. Dots are users colored by network, stars mark the network managers, and dashed lines connect networks within the neighbor threshold $\Gamma = 500$ m. The asymmetry in user counts and neighborhood sizes is the structural origin of the non-IID client experience.

We use value-based reinforcement learning, in which the agent estimates $Q_\pi(s, a) = \mathbb{E}_\pi[\sum_{i \geq 0} \gamma^i r(t+i+1) \mid S(t) = s, A(t) = a]$ with discount $\gamma \in (0, 1)$, approximates it by a network $Q(s, a; w)$, and fits it by minimizing the temporal-difference (TD) error [16].

Standard deep Q -networks form the bootstrapped target with a delayed copy w^- of the parameters. Such a target network is a poor fit for a federated system, because the delayed copy is a second piece of state that would have to be either synchronized with the aggregation schedule or kept private and thus inconsistent across clients. We therefore build the learner on DeepMellow [8], which replaces the hard maximum by the mellowmax operator [9],

$$\text{mm}_\omega(Q(s, \cdot; w)) = \frac{1}{\omega} \log \left(\frac{1}{K} \sum_{k=1}^K e^{\omega Q(s, k; w)} \right), \quad (7)$$

a smooth surrogate recovering the mean as $\omega \rightarrow 0$ and the maximum as $\omega \rightarrow \infty$. Because mellowmax is a non-expansion and softens the maximization bias that motivates the target network in the first place, stable learning is attained *without* a delayed copy—which is precisely what makes the client state in our framework a single parameter vector. The target of client n is therefore

$$Y^n(t) = r^n(t+1) + \gamma (1 - d(t+1)) \text{mm}_\omega(Q(s^n(t+1), \cdot; w)), \quad (8)$$

computed from the online parameters under a stop-gradient, with d the terminal indicator. With $e^n(t) = Y^n(t) - Q(s^n(t), a^n(t); w)$, the per-sample loss is the Huber loss $L_\delta(e)$, whose linear tail limits the influence of the rare large TD errors that a non-stationary multi-agent environment inevitably produces. The value network is a three-layer residual multilayer perceptron with 128 units per layer, Leaky-ReLU activations of slope 0.2, an identity output layer of width K , and PyTorch’s default Kaiming-uniform initialization [17]; it holds $d_w = 36,234$ parameters for $K = 10$, and is common to all methods compared in Section IV.

B. Observation Space, Action Space, and Exploration

The state concatenates the quality vector (5) with a binary encoding of the channel currently occupied,

$$s^n(t) = [\text{CBR}^n(t); \text{QV}^n(t)] \in \mathbb{R}^{B+K}, \quad B = \lfloor \log_2 K \rfloor + 1, \quad (9)$$

where $\text{CBR}^n(t) \in \{0, 1\}^B$ is the channel binary representation of the previous allocation; including it is what allows the agent to distinguish “stay” from “switch” and hence to learn the spectrum-mobility trade-off. The action space is the channel set $\mathcal{K}$, so the output layer has one value per channel.

Two mechanisms shape action selection. First, channels on which no user would meet the QoS target are removed by invalid-action masking, $\tilde{Q}(s^n, k; w) = -\infty$ if $\text{QV}_k^n = 0$ and $Q(s^n, k; w)$ otherwise; in the degenerate case $\text{QV}^n = \mathbf{0}$ no decision is taken and the current channel is retained. Masking prevents wasted exploration on provably unusable channels and reduces collisions. Second, exploration follows a Boltzmann policy over the masked values,

$$\Pr(a^n(t) = k \mid s^n(t)) = \frac{e^{\beta \tilde{Q}(s^n(t), k; w)}}{\sum_{k' \in \mathcal{K}} e^{\beta \tilde{Q}(s^n(t), k'; w)}}, \quad (10)$$

sampled with probability ϵ_b and replaced by the greedy choice otherwise. We anneal ϵ_b linearly from 0.5 to 0.01 over the first half of training and hold it fixed thereafter. Note that ϵ_b is a global schedule indexed by the communication round: all clients in a round explore at the same rate, which keeps the cohort’s data collection statistically comparable.

C. Reward Design and the Locality of the Learning Signal

The reward must express two competing goals: occupy a high-quality channel, and stop moving once a good allocation has been found. The personal component r_p of Algorithm 1 does both. It grants a large fixed reward r_{desired} when the fraction of satisfied users on the chosen channel exceeds a threshold ζ ; otherwise it returns the rank of the chosen channel among all K channels, mapped to $[-1, 1]$, so that selecting the best available channel is rewarded even when no channel is outright good. A multiplicative bonus $c_1 > 1$ is applied when the agent keeps its previous channel, penalizing needless spectrum mobility. With $\zeta = 0.9$, $r_{\text{desired}} = 4$, $c_1 = 1.1$ and $T = 20$ decision points per episode, the maximum attainable accumulated reward is $T c_1 r_{\text{desired}} = 88$, which we use in Section IV as an absolute reference.

A social-welfare component r_{sw} makes each agent accountable for the externalities it imposes on its neighbors. With the interference neighborhood $\mathcal{B}^n = \{m \neq n : \|\mu^m - \mu^n\|_2 \leq \Gamma\}$, it is the mean personal reward collected by those neighbors between two consecutive decisions of network n (zero if $\mathcal{B}^n = \emptyset$), and the total reward is the convex combination

$$r = \rho r_p + (1 - \rho) r_{sw}, \quad (11)$$

with personal-reward fraction $\rho \in [0, 1]$ trading self-interest against cooperation.

This construction deserves attention in a federated context, because it is the one place where an agent’s reward depends on quantities computed by others: evaluating r_{sw} requires

Algorithm 1 Personal reward r_p of network $\mathcal{N}^n$

Require: $QV^n(t)$, actions $a^n(t)$, $a^n(t-1)$
Ensure: r_p
1: $v \leftarrow QV_{a^n(t)}^n(t)$
2: **if** $v \geq \zeta$ **then**
3: $r_p \leftarrow r^{\text{desired}}$
4: **else**
5: $SQV^n \leftarrow \text{sort}(QV^n(t)); i \leftarrow 0$
6: **while** $i \leq K$ **and** $v \geq SQV_i^n$ **do**
7: $i \leftarrow i + 1$
8: **end while**
9: $r_p \leftarrow 2(i/K - 0.5)$ $\triangleright$ rank of the chosen channel
10: **end if**
11: **if** $a^n(t) = a^n(t-1)$ **and** $r_p > 0$ **then**
12: $r_p \leftarrow c_1 r_p$ $\triangleright$ stability bonus
13: **end if**

network n to learn the personal rewards obtained by its neighbors, an exchange of information during *training* that is modest but real. Setting $\rho = 1$ removes the term altogether and makes the learning signal strictly local—an agent then needs nothing beyond its own quality vector and action history. We adopt $\rho = 1$ in CARLTON FL for exactly this reason, and retain $\rho = 0.7$ for the centralized baseline, which already assumes a coordinator pooling all trajectories and for which the social term is therefore free. Our framework thus operates under a *stricter* information constraint than the baseline it is compared against, making the comparison conservative rather than favorable; Section IV reports a control experiment isolating ρ .

D. Federated Training Procedure

Training proceeds over B communication rounds, summarized in Algorithm 2. In round t the server **broadcasts** w^t to all participating clients. A scenario with N_t networks is then generated and played for $T N_t$ steps (**interaction**), so that every client obtains exactly T decision points and appends the corresponding transitions to its own persistent memory $\mathcal{D}^n$. Each client then performs E gradient steps on mini-batches of size b_z drawn from $\mathcal{D}^n$ alone (**local update**); clients holding fewer than b_z transitions skip the update and return the broadcast model unchanged, which occurs only in the earliest rounds. Finally the clients upload $w_n^{t,E}$ and the server **aggregates** them into w^{t+1} .

Two features distinguish this loop from textbook federated learning. First, the dataset is not fixed: clients *generate* their own data with the policy they are currently learning, so $\mathcal{D}^n$ grows and its distribution shifts as training progresses. Persistence is what makes this workable—a buffer retained across rounds preserves experience gathered under earlier policies and topologies, which both stabilizes the TD targets and prevents a client from overfitting the single episode it has just played. Second, the cohort is redrawn each round, since N_t is resampled and a fresh topology generated. This is structurally identical to partial client participation, and it has the useful side effect that the global model is never allowed to specialize to one deployment geometry.

Local objective. Let $F_n(w) = \mathbb{E}_{(s,a,r,s') \sim \mathcal{D}^n} [L_\delta(e(s,a,r,s'); w)]$ be the expected TD loss at client n . In FEDAVG each client simply applies E stochastic gradient steps to F_n . This is the natural baseline, and when E is small it is a good one; but the whole point of many local steps is to reduce the number of communication rounds, and as E grows each client converges toward the minimizer of its *own* objective. Under the heterogeneity of Section II-C those minimizers are far apart and their average need not be good for any of them—the client-drift phenomenon. FEDPROX [18] addresses it by replacing the local objective with the proximal surrogate

$$h_n(w; w^t) = F_n(w) + \frac{\mu}{2} \|w - w^t\|_2^2, \quad (12)$$

whose gradient adds a restoring force $\mu(w - w^t)$ toward the broadcast model at every local step. The penalty does not forbid local progress; it bounds how far a client may travel from the consensus point within a round, which is precisely the quantity that determines whether averaging is meaningful. Thus μ interpolates between two failure modes: $\mu = 0$ recovers FEDAVG exactly—a property we assert as a numerical regression test on the implementation—while excessively large μ freezes the clients at w^t .

The proximal term is a particularly natural fit here. In supervised federated learning the anchor w^t is merely a consensus point; in value-based reinforcement learning it is additionally the parameter vector that generated the bootstrapped targets (8) at the start of the round, so penalizing displacement from w^t also limits the extent to which each client chases a target it is itself moving—an off-policy instability that is independent of federation but amplified by long local phases.

E. Aggregation and Why Federation Helps

The server combines the uploaded models by weighted averaging, $w^{t+1} = \sum_{n \in \mathcal{C}_t} (n_k / \sum_{j \in \mathcal{C}_t} n_j) w_n^{t,E}$, where $\mathcal{C}_t$ is the cohort of round t and n_k the number of transitions contributed by client k . Because the horizon is fixed at $T N_t$ steps, every client collects exactly T new transitions per round, so the weights are uniform and aggregation reduces to the plain mean. This is a consequence of the round-robin turn structure of the environment, not an assumption. Aggregation is performed independently per parameter tensor and is therefore agnostic to the architecture.

Since all clients start the round from the same w^t , it is convenient to write the update in terms of the local displacements $\Delta_n^t = w_n^{t,E} - w^t$, giving $w^{t+1} = w^t + |\mathcal{C}_t|^{-1} \sum_n \Delta_n^t$. In this form the mechanism by which federation helps is explicit: the global step is the mean of $|\mathcal{C}_t|$ displacements computed from independently sampled topologies and independently sampled mini-batches. Components of Δ_n^t reflecting a client's idiosyncratic interference neighborhood are attenuated by averaging, while components shared across clients survive. Periodic averaging thus acts as a variance-reduction mechanism on top of, and orthogonal to, mini-batch averaging—which is why we expect, and in Section IV observe, federated training not merely to tolerate the loss of centralized experience but to benefit from its absence.

Algorithm 2 CARLTON FL training procedure

Require: Episodes B , local training steps E , proximal weight μ , batch size b_z , buffer capacity S_z , schedule $\{\epsilon_b(t)\}$

Ensure: Global policy parameters w^*

```

1: Server initializes  $w^0$ ; each network manager  $n$  initializes
   an empty persistent  $\mathcal{D}^n$  of capacity  $S_z$ 
2: for  $t = 0, \dots, B - 1$  do
3:   Sample  $N_t$  with  $N_t < K$ , generate a scenario, and
   sample participating clients  $\mathcal{C}_t = \{1, \dots, N_t\}$ 
4:   Broadcast:  $w_n^{t,0} \leftarrow w^t, \forall n \in \mathcal{C}_t$ 
5:   for  $\tau = 1, \dots, T$  do ▷ interaction phase
6:     for  $n \in \mathcal{C}_t$  in turn order do
7:       Collect BSINR reports; form  $QV^n$  by (5) and
        $s^n(\tau)$  by (9)
8:       Sample  $a^n(\tau)$  by (10) with  $\epsilon_b(t)$ ; observe  $r^n$ ,
        $s^n(\tau+1)$ 
9:        $\mathcal{D}^n \leftarrow \mathcal{D}^n \cup \{(s^n(\tau), a^n(\tau), r^n, s^n(\tau+1))\}$ 
10:    end for
11:  end for
12:  for  $n \in \mathcal{C}_t$  do ▷ local update, in parallel
13:    if  $|\mathcal{D}^n| \geq b_z$  then
14:      for  $e = 1, \dots, E$  do
15:        Sample a mini-batch of size  $b_z$  from  $\mathcal{D}^n$ 
16:         $w_n^{t,e} \leftarrow w_n^{t,e-1} - \eta \nabla [\hat{F}_n + \frac{\mu}{2} \|w_n^{t,e-1} -$ 
        $w^t\|^2]$ 
17:      end for
18:    else
19:       $w_n^{t,E} \leftarrow w^t$ 
20:    end if
21:    Upload  $w_n^{t,E}$ 
22:  end for
23:  Aggregate:  $w^{t+1} \leftarrow \frac{1}{|\mathcal{C}_t|} \sum_{n \in \mathcal{C}_t} w_n^{t,E}$ 
24: end for
25: return  $w^* \leftarrow w^B$ 

```

F. Cost, Execution, and Privacy

It is worth quantifying what the change of paradigm costs and saves, since the answer is not uniformly favorable. *Downlink* is unchanged: both paradigms broadcast the model to every manager before each episode. *Uplink* differs in kind. CTDE uploads T raw transitions of $2(B + K) + 3$ scalars each, about 2.5 kB per client per round, whereas CARLTON FL uploads $d_w = 36,234$ parameters, about 145 kB—roughly $58\times$ more for a model this compact. We state this plainly: federated training is not, here, a bandwidth optimization. What it buys is that the payload is *fixed in size and content-independent*: it does not grow with the number of users M^n , with episode length, or with buffer retention, and it carries no measurement attributable to a specific user. Where uplink is binding, the standard remedies apply unmodified—8-bit quantization divides the cost fourfold, and aggregating every R rounds divides it by R . *Computation* moves in the opposite direction: instead of all gradient steps running on one node whose load grows with the deployment, each client performs E steps on its own hardware concurrently, at cost $O(E b_z d_w)$ independent of N_t , and a client that drops out simply does not

contribute to the average, whereas the loss of a central learner halts CTDE training entirely.

Once training terminates, w^* is distributed to the managers and the server is discarded. Execution is fully distributed and message-free: at each decision instant a manager collects the BSINR reports of its own users, forms QV^n , applies the mask, and selects $\arg \max_k \hat{Q}(s^n, k; w^*)$ with a single forward pass. The federated construction thus changes the training phase only, leaving intact the deployment properties that motivate distributed DCA in the first place.

Regarding privacy, our claim is precise and deliberately limited. What is eliminated is the export of raw sensing data—the per-user BSINR reports, the quality vectors, the occupied-channel sequence, and the rewards—which are exactly the quantities from which an observer of a global replay memory could reconstruct a network’s spatial situation, load, and traffic pattern. What remains exposed is the parameter update Δ_n^t , and model deltas are known to leak information about the data that produced them [14]. We therefore do not claim formal privacy, but note that CARLTON FL is compatible without algorithmic modification with the standard defenses: secure aggregation reveals only the sum to the server, and differentially private updates follow from clipping Δ_n^t and adding calibrated noise. Quantifying the resulting privacy–utility trade-off is left to future work.

IV. EXPERIMENTS AND DISCUSSION

In this section we present the experimental study of CARLTON FL. We begin with the simulation environment and the evaluation protocol in Subsection IV-A, and define in Subsection IV-B the physical performance measures used throughout, which are the ones introduced for the centralized state of the art in [1]. Subsection IV-C analyzes the training behavior of CARLTON FL, and Subsection IV-D provides a performance comparison against CARLTON and the alternative algorithms evaluated in [1].

A. Simulation Environment and Evaluation Protocol

The simulation environment follows Section II-A: a total band split into $K = 10$ overlapping channels with carriers spaced 2 MHz apart over 208–226 MHz, Egli propagation, the spectral attenuation schedule of Section II-A, and a thermal noise floor of $I_T = -104.9$ dBm. In every scenario instance the number of networks N is drawn at random, each network serves $M^n \sim U\{1, \dots, 15\}$ users scattered around its center by a bivariate Gaussian with covariance $\text{diag}(50^2, 50^2)$ m², and centers are placed sequentially at a random radius $r \sim U[50, 500]$ m and angle $\theta \sim U[0, 2\pi]$ from a previously placed center. Each network is managed by the user minimizing the total distance to all others, and every network receives exactly $T = 20$ decision points, so the scenario horizon is TN steps. Fig. 3 illustrates a representative instance.

Training scenarios are restricted to $N \leq 7$ networks, as in [1]; scenarios with $N > 7$ are therefore never seen during training and are used to probe generalization. Every run comprises $B = 1000$ rounds (episodes for the centralized baselines) followed by 100 frozen-weight inference episodes,

with ϵ_b annealed from 0.5 to 0.01 over the first half of training. The value network is the 36,234-parameter residual MLP of Section III-A, and all remaining hyperparameters follow Table II of [1]. We report four methods over 30 independent seeds each:

- CARLTON (PyTorch CTDE): centralized training with a shared global replay memory and $\rho = 0.7$;
- *TF CTDE*: TensorFlow centralized baseline under the same reward ($\rho = 0.7$);
- FEDAVG: federated training with $E = 10$ local steps and $\mu = 0$;
- FEDPROX: the same with proximal weight $\mu = 0.01$.

The pair $(E, \mu) = (10, 0.01)$ is the outcome of a dense sweep over $E \in \{1, 2, 5, 10, 15, 20, 40, 60, 80, 120, 200, 300\}$ and a prior sweep over $\mu \in \{0, 0.001, 0.01, 0.1\}$ (Section IV-C). Federated clients use personal reward only ($\rho = 1$) and never observe a neighbor's reward, whereas the centralized baselines retain the social term ($\rho = 0.7$); the federated setting is thus the stricter one from an information standpoint. Following the protocol of [1], trained policies are finally evaluated on 30 games for each $N \in \{2, \dots, 15\}$, i.e., 420 games per algorithm.

B. Performance Metrics

We adopt the physical performance measures defined in [1], so that our results are directly comparable with the reported behavior of the centralized state of the art. Let $c^n(t)$ denote the channel occupied by network $\mathcal{N}^n$ at step t , and recall the quality vector $\mathbf{QV}^n$ of (5). At the end of a scenario the *channel quality* (CQ) vector collects the quality each network actually experiences on the channel it settled on,

$$\begin{aligned} \text{CQ} &= [\mathbf{QV}_{c^1(T)}^1(T), \dots, \mathbf{QV}_{c^N(T)}^N(T)], \\ \overline{\text{CQ}} &= \mathbb{E}[\text{CQ}], \quad \min_{\text{CQ}} = \min(\text{CQ}). \end{aligned} \quad (13)$$

The mean $\overline{\text{CQ}}$ measures aggregate spectral health, while $\min_{\text{CQ}}$ measures fairness: it is the quality of the worst-served network, and it is the term that a purely self-interested allocation tends to sacrifice. Their average, $\mathbb{E}[(\text{CQ} + \min_{\text{CQ}})/2]$, is the composite reported in Figs. 13 and 15 of [1] and is our primary metric.

Spectrum mobility is captured by the average number of channel changes until convergence per network (ANCC), normalized into a score

$$\text{ANCCS} = 1 - \frac{\text{ANCC}}{T}, \quad (14)$$

and by the convergence time (CT), defined as the step of the last channel change performed by any network in the scenario, with score

$$\text{CTS} = 1 - \frac{\text{CT}}{T \cdot N}. \quad (15)$$

Both matter operationally: each channel change costs signaling and a brief service interruption, and a slow-converging allocation is useless in a band whose occupancy changes on the same timescale.

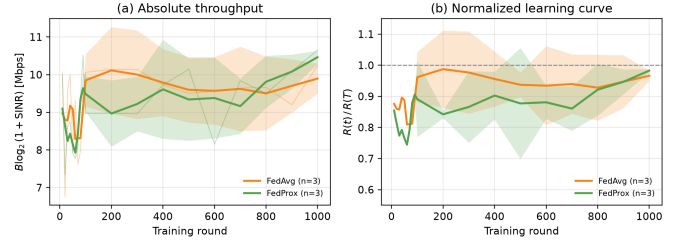


Fig. 4. Shannon throughput of the learned allocation. (a) Absolute rate $B \log_2(1 + \text{SINR})$ in Mbps (light smoothing; faint lines show the raw mean). (b) Normalized learning curve $R(t)/R(T)$, where T is the final training round. Mean $\pm$ std over 3 seeds; $B = 2$ MHz.

The third measure is the spectrum efficiency score (SES), which quantifies how much clean spectrum the converged allocation leaves behind for a network that might join later,

$$\text{SES} = \mathbb{E}[\Psi] = \frac{1}{N} \sum_{n=1}^N \Psi^n, \quad \Psi^n = \left(\sum_{k=1}^K (\mathbf{QV}_k^n(T))^2 \right)^{0.5} / K^{0.5}. \quad (16)$$

A high SES indicates that the incumbents converged to a solution that distorts few of the remaining channels.

Finally, these quantities are combined into the linear weighted score

$$\text{WS} = w_1 \overline{\text{CQ}} + w_2 \text{ANCCS} + w_3 \text{CTS} + w_4 \text{SES}, \quad (17)$$

with $(w_1, w_2, w_3, w_4) = (0.4, 0.1, 0.4, 0.1)$ as in [1]. The larger weights are placed on $\overline{\text{CQ}}$ and CTS because sustained channel quality determines whether the QoS-SINR constraint of (4) is met at all, and fast convergence determines whether the allocation is still valid by the time it is reached; ANCCS and SES receive smaller weights since reduced mobility and residual spectrum cleanliness are desirable but secondary to meeting QoS quickly.

C. CARLTON FL's Training Dynamics

We track the physical rate that the learned allocation actually delivers, rather than treating the DeepMellow surrogate reward as the primary figure of merit. For each saved global checkpoint we freeze the weights, play a fixed set of in-sample scenarios, and convert the SINR each network achieves on its final channel into a Shannon rate

$$R^n = B \log_2(1 + \text{SINR}^n), \quad B = 2 \text{ MHz}, \quad (18)$$

averaged over the users of network $\mathcal{N}^n$. Fig. 4 reports the network-mean rate as a function of the training round for FEDAVG and FEDPROX (mean $\pm$ std over three seeds). Panel (a) shows the absolute Shannon throughput in Mbps; panel (b) normalizes each curve by its final-round value $R(T)$, so that the learning dynamics are visible on a common scale. Both methods climb toward their operating point within the first few hundred rounds and then stabilize, confirming that the federated policies translate improving value estimates into higher delivered throughput.

The number of local steps E between aggregations is the main free parameter of Algorithm 2. Fig. 5 sweeps

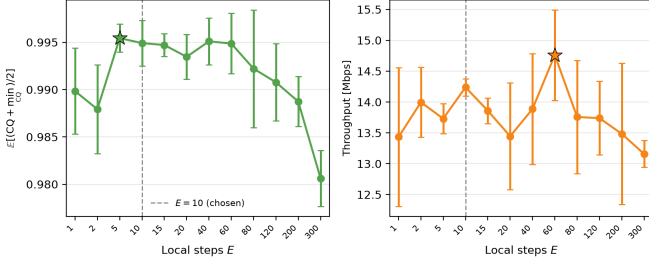


Fig. 5. Sensitivity of FEDPROX ($\mu = 0.01$) to the number of local steps E over the dense grid $\{1, 2, 5, 10, 15, 20, 40, 60, 80, 120, 200, 300\}$. Left: composite channel quality $E[(CQ + \min_{CQ})/2]$. Right: Shannon throughput [Mbps]. The dashed line marks the chosen operating point $E = 10$; the star marks the empirical maximum of each panel.

$E \in \{1, 2, 5, 10, 15, 20, 40, 60, 80, 120, 200, 300\}$ for FEDPROX ($\mu = 0.01$) and scores the final global model of each arm on composite channel quality and Shannon throughput. The operating point used throughout the rest of the paper is $E = 10$, selected as the best trade-off on this dense grid; both much shorter and much longer local phases lose measurable quality, consistent with the bias–variance trade-off that FEDPROX is designed to regularize.

Two remarks are important for interpretation. First, the DeepMellow training reward is a surrogate used only during learning; the operational objectives of interest are CQ, $\min CQ$, WS, and the Shannon rate of (18) (Section IV-D). Second, the federated runs use $\rho = 1$ and therefore never observe neighbors’ personal rewards during learning, whereas CTDE benefits from the social term ($\rho = 0.7$). Under that stricter constraint, matching or approaching—not exceeding—CTDE on channel-quality metrics is the success criterion.

D. Performance Comparison

We now compare CARLTON FL against the main benchmark algorithms for this model. As an upper bound, we include a *centralized graph-coloring* allocator, extended to overlapping channels by grouping channels into interference-compatible sets and solving the resulting assignment with an iterative local search; it is centralized and therefore not implementable in a distributed system. The distributed comparisons are: (i) CARLTON itself (PyTorch CTDE), with and without the post-processing threshold φ that suppresses a channel switch unless it improves CQ by at least φ ; (ii) the Jamming Avoidance Response (JAR) heuristic as implemented in [1], a bio-inspired scheme that shifts the carrier by ± 2 MHz only when the move improves CQ by at least 0.05, which is the state of the art for this model among non-learning methods; (iii) the multi-agent DDQN allocator MADDQN-DCA [5]; (iv) the deep recurrent RL scheme IDSA [6]; and (v) the Random Agent (RA) [12], which draws an initial channel and never switches. All algorithms are assessed on the same 420-game sweep (30 games for each $N \in \{2, \dots, 15\}$). Figs. 6–7 summarize the outcome.

The in-sample result is the central one. Fig. 6 shows the PyTorch CTDE baseline at 0.983, with FEDAVG and FEDPROX at 0.982 and 0.978—matching the centralized learner to within

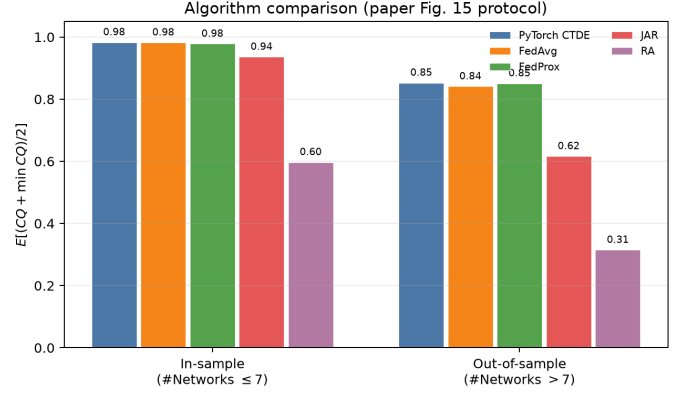


Fig. 6. In-sample versus out-of-sample composite channel quality $E[(CQ + \min_{CQ})/2]$. In-sample: $N \leq 7$; out-of-sample: $N > 7$.

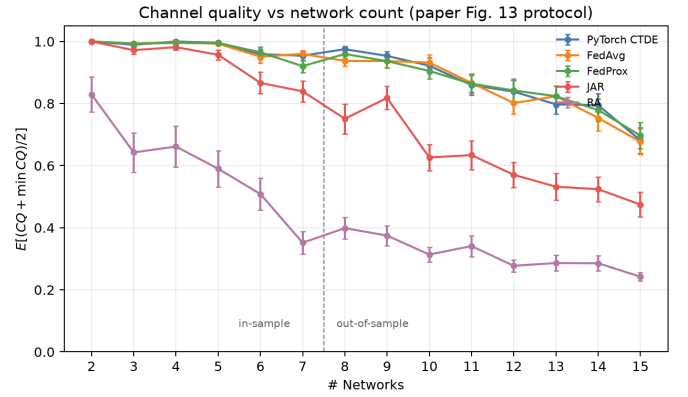


Fig. 7. Composite channel quality versus number of networks. The dashed line marks the training support $N \leq 7$.

the metric resolution, and matching the 0.98 reported for CARLTON and the 0.99 graph-coloring bound in [1]. Federated training therefore keeps pace with pooled-experience CTDE on the domain it was trained on, while never pooling raw sensing data and while learning from a strictly local reward ($\rho = 1$). This is the expected best case: aggregating parameters instead of experience cannot create information that pooled training does not already have, so staying within a fraction of a percent of CTDE—not surpassing it—is the meaningful outcome.

Fig. 7 resolves the same comparison as a function of N . CTDE leads by a thin margin across most of the sweep, and both federated methods stay well above JAR and RA for every network count. Out-of-sample ($N > 7$) the all-sample gap remains small: CTDE attains a composite of 0.853 against 0.851 for FEDPROX and 0.841 for FEDAVG. Federation thus retains the same operating regime as CTDE under generalization to network counts never seen in training, without claiming an information-theoretic advantage over the centralized learner.

The remaining metrics of Subsection IV-B complete the picture. Averaged over the sweep, CTDE, FEDAVG, and FEDPROX all attain WS near 0.931, above 0.910 for JAR and 0.818 for RA. Their convergence-time scores (0.931–0.933) sit just below JAR (0.945) and RA (0.960)—both

of which converge quickly only because they barely move. CARLTON FL therefore reaches near-CTDE channel quality without paying for it in spectrum mobility, and without needing the post-processing threshold φ that [1] introduces to recover convergence time.

Finally, we caution against reading differences in the DeepMellow training reward as evidence that federation is the stronger learner. Federated and centralized runs optimize that surrogate under different ρ ; the operational claim of this paper rests on Figs. 6–7, where CTDE (CARLTON) is slightly ahead overall and CARLTON FL remains competitive under a strictly stronger privacy constraint.

V. CONCLUSION

We presented CARLTON FL, a federated deep reinforcement learning framework for SINR-aware distributed dynamic channel allocation under inter-carrier interference. By replacing the global replay memory of CTDE with private, persistent per-client buffers and aggregating only model parameters via FEDAVG or FEDPROX, the framework removes the need to export raw sensing histories while retaining a fully decentralized execution phase. Each network manager thus acts as a closed-loop agent with private memory and parameter-only coordination—a privacy-preserving instance of multi-agent edge intelligence for spectrum management. On the channel-quality metrics $\mathbb{E}[(\text{CQ} + \min \text{CQ})/2]$ and WS, evaluated over 420 games per method against RA, JAR, and a PyTorch CTDE (CARLTON) baseline, the centralized learner remains slightly ahead (0.909 versus 0.905 all-sample composite for FEDPROX; in-sample 0.983 versus 0.978), while CARLTON FL stays in the same regime without sharing raw sensing data. Federation is therefore a practical path to data sovereignty in this setting: it does not claim to outperform CARLTON, but trades a thin margin against CTDE for a strictly stronger privacy constraint.

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